Bryan Lai

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EDUCATION

GEORGIA INSTITUTE OF TECHNOLOGY — **Atlanta, GA**
M.S. in Computer Science, Specialization in Human-Computer Interaction ***GPA: 3.9/4.0*** — **August 2024 - August 2026**

STONY BROOK UNIVERSITY — **New York, NY**
B.S in Computer Science; B.S in Applied Mathematics & Statistics — **August 2020 - December 2023**

PROFESSIONAL EXPERIENCE

ALTRO HEALTH — **REMOTE**
Full Stack Engineer (L2) — **Sep 2026 - Present**

- Built features for internal clinical operations tools in Next.js and MongoDB on Google Cloud, streamlining day-to-day workflows for the clinical team.
- Helped launch mobile phlebotomy as a new lab offering, building the end-to-end flow on FHIR-based data models for lab orders and patient records with Junction API.
- Shipped to production within the first month as part of a five-person engineering team in a HIPAA-regulated healthcare environment.

SOTER TECHNOLOGIES — **RONKONKOMA, NY**
Software Engineer — **July 2024 - Sep 2026**

- Designed multiple interfaces, UML diagrams, and GUIs for an internal shipping website for vape devices, including login pages and client information, utilizing ReactJS, MUI, and MySQL, leading to a 20% increase in user engagement.
- Implemented a feature leveraging Microsoft Azure API web services to enable Microsoft work/school account logins on a platform that provides vape alerts, allowing users to seamlessly retrieve, create, and delete account data.
- Developing healthcare systems (eMedNY) with a focus on token-based authentication, ensuring secure access and data protection by implementing JWT validation and handling token expiration across multiple services.

Support Engineer — **April 2024 - June 2024**

- Oversaw the entire IT support department, providing consulting, technical support, and problem tracking services to clients via CRM systems (Zoho, Confluence, and Jira), resulting in a customer rating of 92%.

Research & Development Intern — **May 2022 - Aug 2022**

- Coordinated with a team of three to develop a healthcare social media platform connecting hospital patients with their loved ones.
- Reduced costs by 25% through a strategic switch to a different messaging API service, improving mobile device authentication.
- Authored technical documentation for Flysense's API, providing essential information for new customers.
- Engineered a mobile app facilitating seamless integration with vaping detection software and security APIs, enabling instant push notifications.

RELEVANT COURSEWORK

- **CS 6250: Big Data in Healthcare:** Coursework in scalable machine learning and big data systems (Hadoop, Spark, Graph DB) applied to healthcare analytics, including predictive modeling, computational phenotyping, and patient similarity.
- **CS 6440: Introduction to Health Informatics:** Hands-on work with the FHIR healthcare interoperability standard and Docker-based containerization for healthcare systems.

PROJECTS

MedTracker

- Designed and implemented a full-stack healthcare tracking application using a FastAPI (Python) backend built on the FHIR standard, with Firebase for authentication and data storage, and a React frontend. Enabled structured storage, redis cache retrieval, and visualization of patient health data to track medication adherence with synthetic data.

Map Redistricting Application

- Collaborated with 3 team members to create a state redistricting application that calculates demographic data using Java SpringBoot, MongoDB, JavaScript, and Python (pandas library). Worked on the front-end on displaying both the map and table of district data.

Spotify Data Visualization Dashboard

- Analyzed trends and correlations with Top 2023 Spotify song attributes using Python and D3.js, utilizing techniques such as MDS correlation, PCA, k-clusters, biplots, parallel coordinates, area charts, and histograms.

TECHNICAL SKILLS

Languages: Java, Javascript, SQL, Python, HTML, CSS, C
Database/Frameworks/Libraries: ReactJS, Node.js, Bootstrap, Pandas, React Native
DevOps/Other Technologies: Docker, Git, AWS, Visual Studio, Azure